



# Computer Vision Final Project

**Team members:** 3 students

## Project Options (Choose One)

### ▪ Option 1: **Smart Shape-Based Object Counter & Analyzer**

In this project, you will build a system that can **detect and count objects** such as cars and bicycles in real-time.

But instead of stopping at detection, you will go deeper and analyze the **shape of each object** using image processing techniques like edges, contours, and Hough Transform.

- **What is the goal?**

- Detect multiple objects in real-time
- Understand the structure of objects using edges and shapes
- Count how many cars, bicycles, etc. appear in the scene
- Improve the results using image processing

- **Requirements:**

1. **Object Detection:**

Make sure the system works with more than one object at the same time

2. **Image Processing:**

Inside each object, apply:

- Grayscale
- Gaussian Blur
- Edge detection: Canny

3. **Shape Detection:**

Use Hough Transform to detect:

- Circles (like wheels)
- Lines

4. **Object Understanding:**

Use the detected shapes to help understand the object:

- Bicycle → usually has two circles
- Car → has lines and circles

## 5. Counting:

Make sure objects are not counted more than once

- Number of cars
- Number of bicycles

## 6. Display:

Show on screen:

- Bounding boxes
- Object names
- Shapes (lines/circles)
- Counters

## 7. Controls:

Add trackbars to control:

- Edge thresholds
- Hough parameters



## ▪ Option 2: **Motion-Aware Smart Surveillance System**

In this project, you will build a system that can **detect moving objects**, track them, and understand their behavior.

The system will work like a simple surveillance camera that analyzes what is happening in the scene.

- **What is the goal?**

- Detect motion in real-time
- Extract moving objects
- Track them across frames
- Count and analyze their movement

- **Requirements:**

- 1. Motion Detection:**

Detect moving areas in the frame.

- 2. Processing:**

- Convert to grayscale
- Apply thresholding
- Use morphological operations to remove noise

- 3. Object Detection:**

- Detect contours of moving objects
- Draw bounding boxes

- 4. Edge Detection:**

For each object apply:

- Canny (required)

- 5. Hough Transform:**

Detect:

- Lines
- Circles

Use them to understand object structure

## 6. Object Classification:

Classify objects based on:

- Size
- Shape
- Movement

Example:

- Small and fast → person
- Large → car

## 7. Counting:

- Count objects entering or leaving the scene
- Use a virtual line or region
- Avoid double-counting

## 8. Display:

- Bounding boxes
- Object type
- Counters

